

Automation with Workflows in MISP

Short version

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MISP Project
https://www.misp-project.org/



Content of the presentation

1. Automation in MISP
2. MISP Workflows
   - Fundamentals
   - Demo with examples
   - Using the system
   - How it can be extended

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Automation in MISP: What already exists?

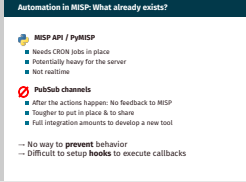
MISP API / PyMISP

- Needs CRON Jobs in place
- Potentially heavy for the server
- Not realtime

PubSub channels

- After the actions happen: No feedback to MISP
- Tougher to put in place & to share
- Full integration amounts to develop a new tool

→ No way to **prevent** behavior
→ Difficult to setup **hooks** to execute callbacks



Simple automation in MISP made easy



- **Visual** dataflow programming
- **Drag & Drop** editor
- Flexible **Plug & Play** system
- **Share** workflows, **debug** and **replay**

Simple automation in MISP made easy
MiSP
Threat Sharing
- **Visual** dataflow programming
- **Drag & Drop** editor
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Example of use-cases

- **Notification** on specifc actions
  - New events matching criteria
  - New users
  - Automated alerts for high-priority IOCs
- **Extend** existing MISP behavior
  - Push data to another system
  - Automatic enrichment
  - Sanity check to block publishing / sharing
  - Curation pipelines
- **Hook** capabilities
  - Assign tasks and notify incident response team members
- ...

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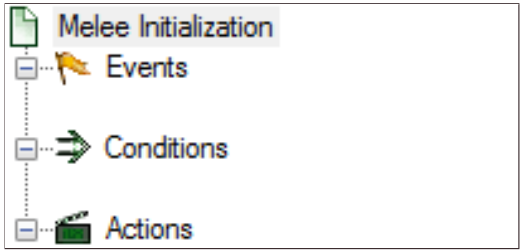
Workflow - Fundamentals

Objective: Start with the foundation to understand the basics



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How does it work



1. An **event** happens in MISP
2. *(optional)* Check if all **conditions** are satisfied
3. Execute all **actions**
   - May prevent MISP to complete its original event

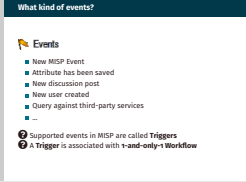
How does it work
Melee Initialization
Events
Conditions
Actions
1. An **event** happens in MISP
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3. Execute all **actions**
   - May prevent MISP to complete its original event

What kind of events?

Events

- New MISP Event
- Attribute has been saved
- New discussion post
- New user created
- Query against third-party services
- ...

❓ Supported events in MISP are called **Triggers**
❓ A **Trigger** is associated with **1-and-only-1 Workflow**



Triggers currently available

Currently 11 triggers can be hooked. 3 being Blocking.

Triggers

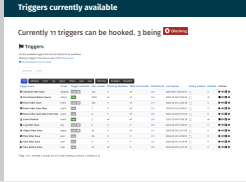
List the available triggers that can be listened to by workflows.
Missing a trigger? Feel free to open a Github issue!
Documentation and concepts

« previous next »

All | attribute | event | log | object | others | post | user | Blocking | Enabled | Disabled

Trigger name	Scope	Trigger overhead	Run counter	Blocking Workflow	MISP Core format	Workflow ID	Last Update	Debug enabled	Enabled	Actions
Attribute After Save	attribute	high	110	✗	✓	160	2023-09-14 06:54:37	☐	✗	
Enrichment Before Query	others	low	2226	✓	✓	162	2023-10-09 07:56:42	☐	✓	
Event After Save	event	high	191	✗	✓	175	2023-10-02 14:55:19	☐	✗	
Event After Save New	event	low	7	✗	✓	182	2023-03-16 14:05:07	☐	✗	
Event After Save New From Pull	event	low	6	✗	✓	183	2023-10-09 07:57:02	☐	✗	
Event Publish	event	low	2	✓	✓	188	2023-10-09 07:56:25	☐	✓	
Log After Save	log	high	0	✗	✗	185	2023-06-05 13:26:50	☐	✗	
Object After Save	object	high	35	✗	✓	161	2023-06-05 13:27:00	☐	✗	
Post After Save	post	low	36	✗	✗	176	2022-07-28 13:59:51	☐	✗	
User After Save	user	low	0	✗	✗	181	2022-08-05 07:19:46	☐	✗	
User Before Save	user	low	42	✓	✗	158	2023-06-05 13:27:25	☐	✗	

Page 1 of 1, showing 1 records out of 11 total, starting on record 1, ending on 11

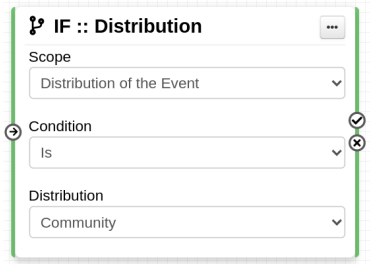


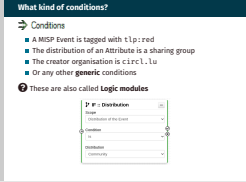
What kind of conditions?

Conditions

- A MISP Event is tagged with `tlp:red`
- The distribution of an Attribute is a sharing group
- The creator organisation is `circl.lu`
- Or any other **generic** conditions

These are also called **Logic modules**



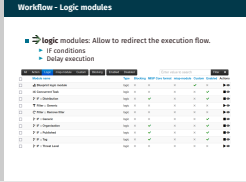


Workflow - Logic modules

- **logic** modules: Allow to redirect the execution flow.
  - IF conditions
  - Delay execution

All | Action | Logic | misp-module | Custom | Blocking | Enabled | Disabled — Enter value to search — Filter

Module name	Type	Blocking	MISP Core format	misp-module	Custom	Enabled	Actions
Blueprint logic module	logic	✗	✗	✗	✓	✗	▶ 👁
Concurrent Task	logic	✗	✗	✗	✗	✓	■ 👁
IF :: Distribution	logic	✗	✓	✗	✗	✓	■ 👁
Filter :: Generic	logic	✗	✗	✗	✗	✗	▶ 👁
Filter :: Remove filter	logic	✗	✗	✗	✗	✗	▶ 👁
IF :: Generic	logic	✗	✗	✗	✗	✓	■ 👁
IF :: Organisation	logic	✗	✓	✗	✗	✓	■ 👁
IF :: Published	logic	✗	✓	✗	✗	✓	■ 👁
IF :: Tag	logic	✗	✓	✗	✗	✓	■ 👁
IF :: Threat Level	logic	✗	✗	✗	✗	✗	▶ 👁

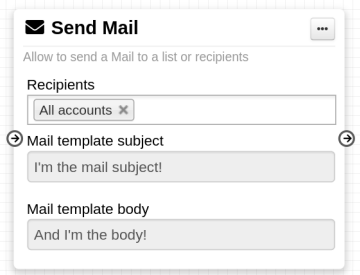


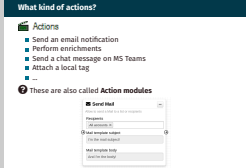
What kind of actions?

Actions

- Send an email notification
- Perform enrichments
- Send a chat message on MS Teams
- Attach a local tag
- ...

❓ These are also called **Action modules**





Workflow - Action modules

- **action** modules: Allow to executes operations
  - Tag operations
  - Send notifications
  - Webhooks & Custom scripts

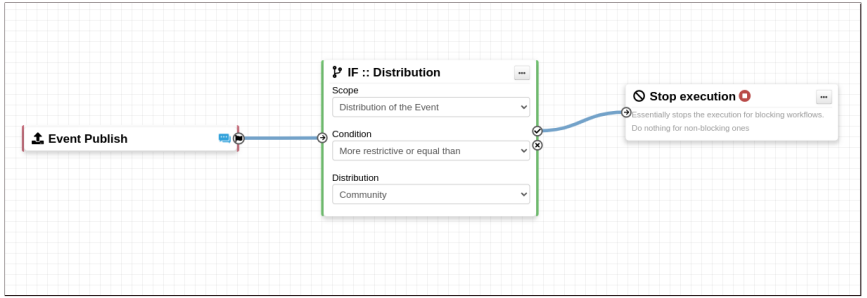
All | Action | Logic | misp-module | Custom | Blocking | Enabled | Disabled | Enter value to search | Filter

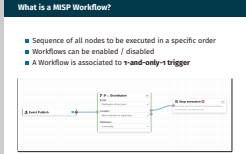
Module name	Type	Blocking	MISP Core format	misp-module	Custom	Enabled	Actions
Attach enrichment	action	×	✓	×	×	✓	■ 👁
Attribute edition operation	action	×	✓	×	×	✓	■ 👁
Attribute IDS Flag operation	action	×	✓	×	×	✓	■ 👁
Blueprint action module	action	×	×	×	✓	✓	■ 👁
Enrich Event	action	×	✓	×	×	✓	■ 👁
mattermost	action	×	×	✓	×	✓	■ 👁
MS Teams Webhook	action	×	×	×	×	✓	■ 👁
Push to ZMQ	action	×	×	×	×	✓	■ 👁
Send Log Mail	action	×	×	×	×	×	▶ 👁
Send Mail	action	×	×	×	×	✓	■ 👁
Splunk HEC export	action	×	✓	×	×	×	▶ 👁
Stop execution	action	✓	×	×	×	✓	■ 👁
Tag operation	action	×	✓	×	×	✓	■ 👁
testaction	action	×	×	✓	×	✓	■ 👁
Webhook	action	×	×	×	×	✓	■ 👁



What is a MISP Workflow?

- Sequence of all nodes to be executed in a specific order
- Workflows can be enabled / disabled
- A Workflow is associated to **1-and-only-1 trigger**





Sources of Workflow modules (0)

Currently 36 built-in modules.

- **Trigger** module (11): built-in **only**
  - Get in touch if you want more
- **Logic** module (10): built-in & **custom**
- **Action** module (20): built-in & **custom**

Sources of Workflow modules (0)
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Sources of Workflow modules (1)

- Built-in **default** modules
  - Part of the MISP codebase
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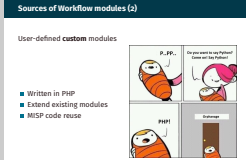


Sources of Workflow modules (2)

User-defined **custom** modules

- Written in PHP
- Extend existing modules
- MISP code reuse

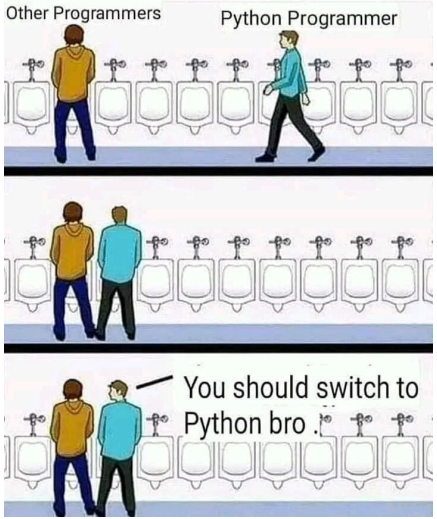


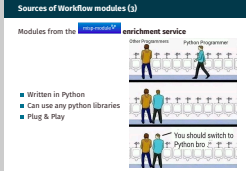


Sources of Workflow modules (3)

Modules from the misp-module **enrichment service**

- Written in Python
- Can use any python libraries
- Plug & Play





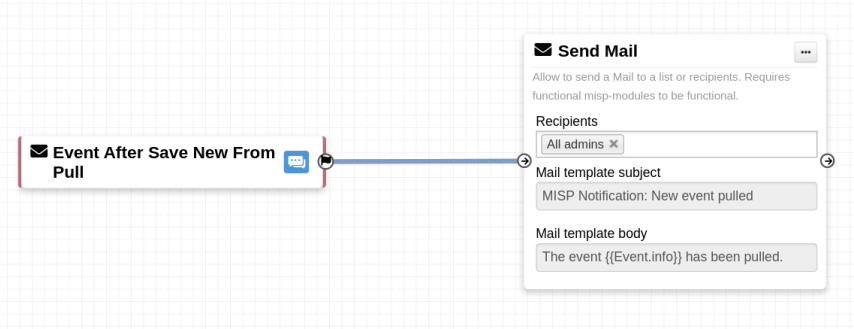
Demo by examples

WF-1. Send an email to **all admins** when a new event has been pulled

WF-2. Block queries on 3rd party services when **tlp:red** or **PAP:red**
- **tlp:red**: For the eyes and ears of individual recipients only
- **PAP:RED**: Only passive actions that are not detectable from the outside

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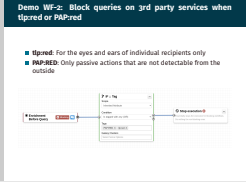
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Everything is ready?

Let’s see how to build a workflow!



Getting started with workflows

Everything is ready?

Let’s see how to build a workflow!

Creating a workflow with the editor

1. Prevent event publication **if tlp:red** tag
   - Send a mail to `admin@admin.test` about potential data leak
2. **else**, send a notification on Mattermost

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Considerations when working with workflows

Objective: Overview of some common pitfalls



Considerations when

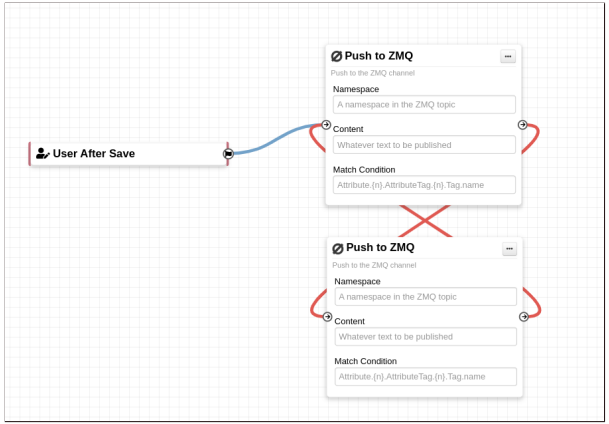
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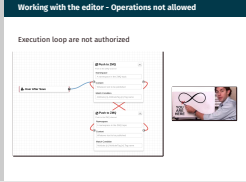


Working with the editor - Operations not allowed

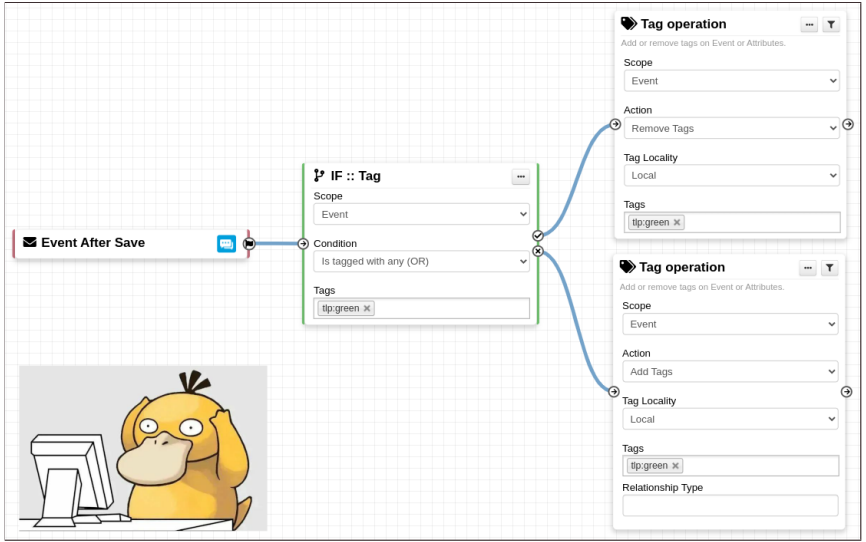
Execution loop are not authorized



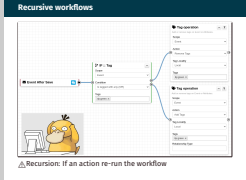




Recursive workflows

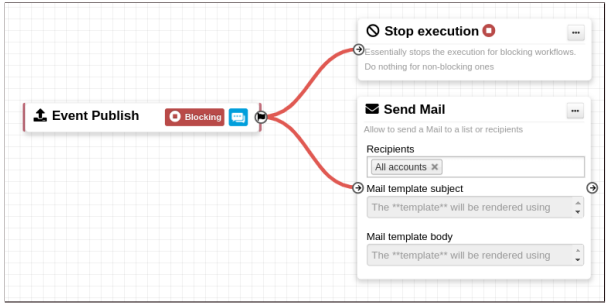


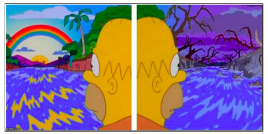
⚠Recursion: If an action re-run the workflow



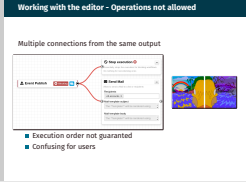
Working with the editor - Operations not allowed

Multiple connections from the same output





- Execution order not guaranted
- Confusing for users



Advanced usage

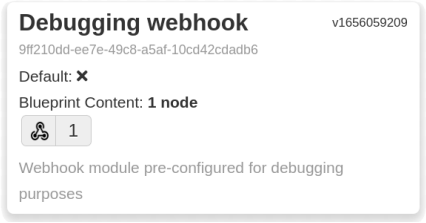
Objective: Overview of Blueprints, Data format and Filtering

Advanced usage

Objective: Overview of Blueprints, Data format and Filtering

Workflow blueprints

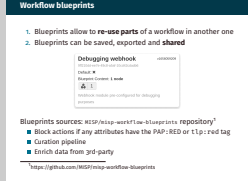
1. Blueprints allow to **re-use parts** of a workflow in another one
2. Blueprints can be saved, exported and **shared**



Blueprints sources: `MISP/misp-workflow-blueprints` repository[1]

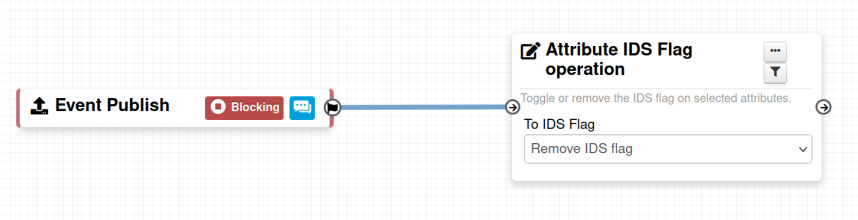
- Block actions if any attributes have the `PAP:RED` or `tlp:red` tag
- Curation pipeline
- Enrich data from 3rd-party

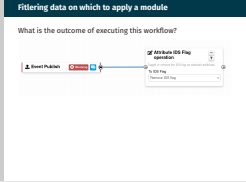
[1] https://github.com/MISP/misp-workflow-blueprints



Fitlering data on which to apply a module

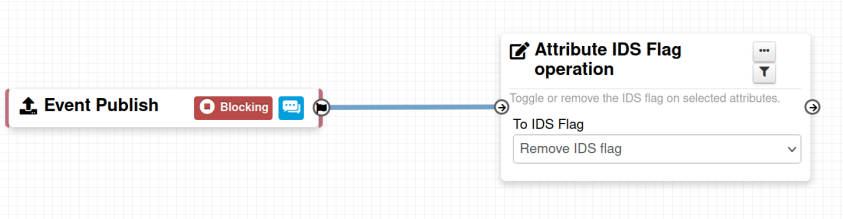
What is the outcome of executing this workflow?





Fitlering data on which to apply a module

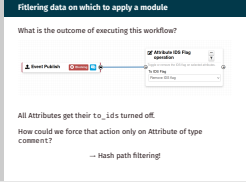
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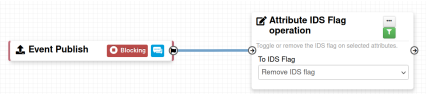
All Attributes get their `to_ids` turned off.

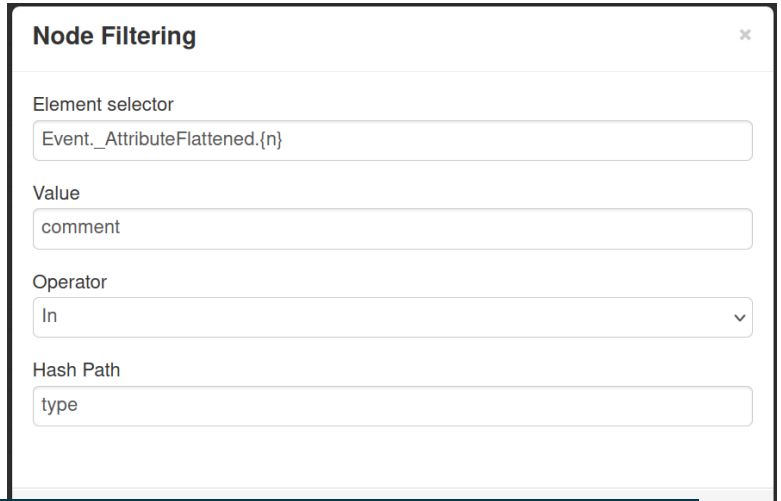
How could we force that action only on Attribute of type `comment`?

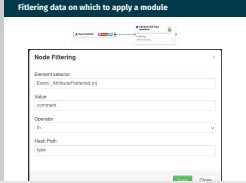
→ Hash path filtering!



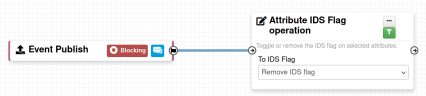
Fitlering data on which to apply a module

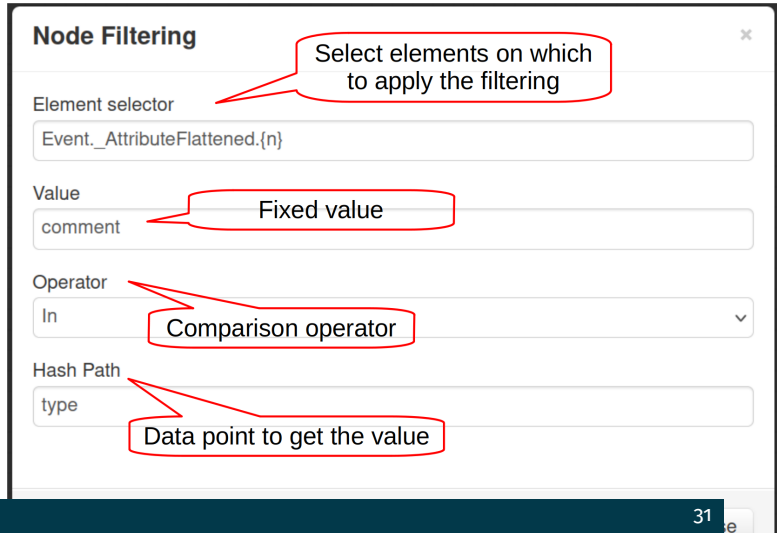


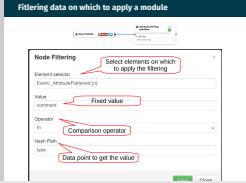




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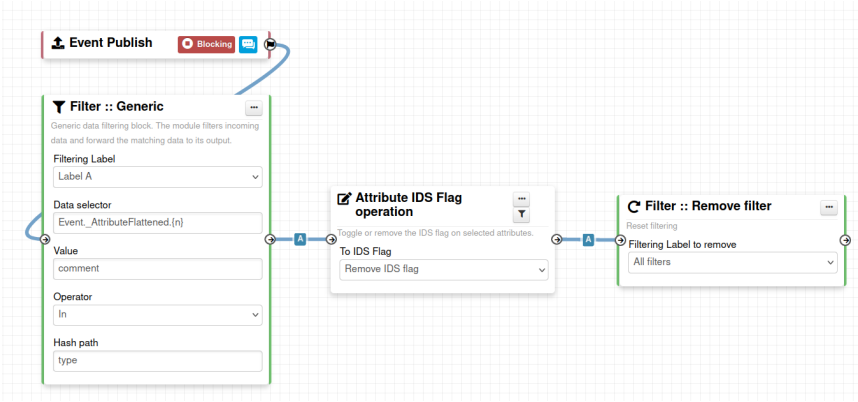


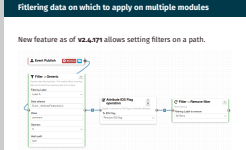




Fitlering data on which to apply on multiple modules

New feature as of **v2.4.171** allows setting filters on a path.





Should I migrate to MISP Workflows

I have automation in place using the API/ZMQ. Should I move to Workflows?

- I have a curation pipeline using the API, should I port it to workflows?
  - **No** in general, but WF can be used to start the curation process or perform simple pre-processing
- What if I want to **block** some actions
  - Put the blocking logic in the WF, keep the remaining outside
- Bottom line is **Keep it simple** for you to maintain

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Future works

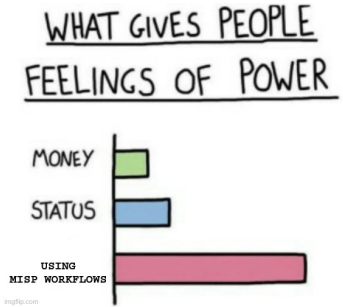
- More modules
- More modules
- More triggers
- Recursion prevention system



Future works
More modules
More modules
More triggers
Recursion prevention system
MISP WORKFLOWS
TODOS

Final words

- Designed to **quickly** and **cheaply** integrate MISP in CTI pipelines
- **Beta** Feature unlikely to change. But still..
- Waiting for feedback!
  - New triggers?
  - New modules?



Final words
Designed to quickly and cheaply integrate MISP in CTI pipelines
Beta Feature unlikely to change. But still..
Waiting for feedback!
New triggers?
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WHAT GIVES PEOPLE FEELINGS OF POWER
MONEY
STATUS